Day-by-Day Work Plan

How Stable Are Regression Conclusions Across Subpopulations?
A Finite-Sample Empirical Study

Project execution schedule

July 16, 2026

1 Overview

This is a compressed work plan for completing the first full draft of the paper under the assumption of approximately **8 hours of focused work per day**. The plan is designed to produce a strong draft in about **12 days**, which is aggressive but realistic if the scope is kept disciplined.

2 Main Objective

The objective is to complete:

- a clear research question,
- a short literature review,
- a reproducible simulation study,
- two real-data case studies,
- all main figures and tables,
- a full paper draft in LaTeX.

3 Daily Work Structure

Each day should roughly follow this structure:

- **2 hours:** reading, planning, and note consolidation,
- **4 hours:** coding, computation, experiments, and debugging,
- **2 hours:** writing the paper or documenting results.

The most important rule is: **write every day**. Do not wait until all results are finished.

4 12-Day Plan

Day 1: Lock the Scope and the Exact Contribution

Main goal: finalize the question, paper scope, and contribution.

Tasks:

(1) Write the exact research question in one sentence, starting from the pooled model as the baseline.

(2) Write a short statement of the paper's contribution.

(3) Decide on the 4 simulation scenarios.

(4) Decide on the 4 regression models to compare.

(5) Decide on the 5 main stability metrics:

- sign stability,
- standardized coefficient drift,
- context-normalized drift,
- transported prediction gap,
- covariate shift summary.

(6) Set up the project folders:

- `data/`
- `scripts/`
- `results/`
- `figures/`
- `paper/`

(7) Create the LaTeX paper file and section headings.

Deliverables by end of day:

- 1-page project memo,
- LaTeX skeleton with sections,
- clear list of models, metrics, and scenarios.

Day 2: Literature Review and Introduction Notes

Main goal: complete the small literature review and map the introduction.

Tasks:

(1) Read and annotate approximately 6–10 relevant papers.

(2) Take notes on:

- subgroup heterogeneity in regression,

  - interaction modeling,
  - transportability / distribution shift,
  - interpretation versus prediction.

(3) Write a rough introduction outline:

  - motivation,
  - prior work,
  - gap,
  - contributions.

(4) Start the bibliography file.

Deliverables by end of day:

- annotated notes for 6–10 references,
- rough introduction draft,
- `references.bib` started.

Day 3: Build the Simulation Engine

Main goal: get one complete simulation pipeline working from start to finish.
Tasks:

(1) Write functions for generating pooled data and subgroup-structured data.

(2) Write functions for fitting:

  - pooled OLS,
  - pooled OLS + group indicator,
  - pooled OLS + interactions,
  - subgroup-specific OLS.

(3) Write helper functions to extract:

  - coefficient estimates,
  - signs,
  - standard errors,
  - RMSE / MAE.

(4) Run one pooled-baseline scenario end-to-end.

(5) Save outputs in a clean format.

Deliverables by end of day:

- working simulation script,
- one successful end-to-end run,
- clear file-saving workflow.

Day 4: Implement All Simulation Scenarios

Main goal: finish the simulation framework for all scenarios.
Tasks:

(1) Implement the 4 main scenarios, starting with the pooled baseline:

- pooled baseline with no heterogeneity,
- covariate shift only,
- coefficient heterogeneity only,
- combined heterogeneity.

(2) Add variation in:

- sample size,
- collinearity,
- subgroup balance.

(3) Run small pilot batches to test all combinations.

(4) Check for bugs and degenerate fits.

Deliverables by end of day:

- stable simulation code for all scenarios,
- pilot outputs confirming the setup works.

Day 5: Run the Full Simulation Study

Main goal: generate the main simulation results.
Tasks:

(1) Run the full simulation grid.

(2) Save all replicate-level results.

(3) Aggregate outputs into summary tables.

(4) Inspect how the pooled baseline behaves under each scenario:

- sign instability,
- standardized coefficient drift,
- context-normalized drift,
- cross-group prediction deterioration,
- covariate shift summary across subgroups.

Deliverables by end of day:

- full raw simulation outputs,
- first summary tables.

Day 6: Create Simulation Figures and Interpret Results

Main goal: turn the simulation output into interpretable figures and findings.

Tasks:

(1) Create 4–6 core plots, such as:
  - sign stability by scenario,
  - standardized coefficient drift by sample size,
  - context-normalized drift by sample size,
  - transported prediction gap by scenario,
  - covariate shift summary by scenario,
  - pooled versus subgroup coefficient comparison plots.

(2) Choose the best visualizations.

(3) Write bullet-point interpretations under each figure.

(4) Decide which results are the strongest and deserve emphasis.

Deliverables by end of day:

- final versions of core simulation figures,
- notes on the main empirical findings.

Day 7: Select and Prepare Real Datasets

Main goal: choose two real datasets and prepare them for analysis.

Tasks:

(1) Choose two datasets with:
  - continuous response,
  - clear subgroup variable,
  - multiple predictors.

(2) Clean and preprocess the data.

(3) Define subgroup variables carefully.

(4) Create exploratory tables and plots.

(5) Check sample sizes within each subgroup.

Deliverables by end of day:

- two analysis-ready datasets,
- initial exploratory summaries.

Day 8: Run the Real-Data Analyses

Main goal: complete the real-data modeling and comparisons.
Tasks:

(1) Fit the pooled model first, then all planned models to both datasets.

(2) Compare:

- pooled coefficients,
- subgroup-specific coefficients,
- interaction terms,
- within-group and out-of-group prediction.

(3) Build 2–4 real-data tables/figures.

(4) Write down how the real-data findings support or contrast with the simulation findings.

Deliverables by end of day:

- completed real-data analyses,
- draft figures/tables for Section 4.

Day 9: Write the Methods Section

Main goal: fully draft the methods section.
Tasks:

(1) Write the notation and problem setup.

(2) Define the models clearly.

(3) Define the stability metrics mathematically.

(4) Describe the simulation design.

(5) Describe the real-data analysis workflow.

Deliverables by end of day:

- full draft of the Methods section.

Day 10: Write the Simulation Results Section

Main goal: write the strongest section of the paper.
Tasks:

(1) Insert the simulation figures and tables into LaTeX.

(2) Write the narrative for each scenario.

(3) Highlight the major patterns:

- when pooled models fail,

- when interaction models help,
- when prediction remains stable but interpretation does not.

(4) Keep interpretation tied directly to the research questions.

Deliverables by end of day:

- full draft of the Simulation Results section.

Day 11: Write the Introduction and Real-Data Section

Main goal: draft the remaining core sections.

Tasks:

(1) Write the Introduction:

- motivation,
- literature review,
- gap,
- contributions.

(2) Write the real-data section.

(3) Make sure the introduction promises only what the paper actually delivers.

Deliverables by end of day:

- draft Introduction,
- draft Real-Data section.

Day 12: Write Discussion, Conclusion, Abstract, and Polish

Main goal: complete the first full draft.

Tasks:

(1) Write the Discussion:

- main findings,
- practical implications,
- limitations,
- future work.

(2) Write the Conclusion.

(3) Write the Abstract last.

(4) Clean the references.

(5) Check notation consistency.

(6) Improve figure captions.

(7) Proofread the full document once from beginning to end.

Deliverables by end of day:

- complete first full draft of the paper.

5 Optional Days 13–14 for Buffer and Revision

If possible, reserve two extra days for revision.

Day 13: Technical Revision

- Re-run any unstable analyses.
- Simplify any overloaded sections.
- Remove weak figures or unnecessary tables.
- Tighten the claims to match the evidence.

Day 14: Writing Revision

- Improve clarity and flow.
- Shorten repetitive paragraphs.
- Make the introduction sharper.
- Make sure the conclusion is specific and not generic.

6 Checklist of Key Deliverables

By the end of the schedule, you should have:

(1) a working bibliography,

(2) a complete LaTeX manuscript,

(3) a simulation pipeline in R,

(4) two real-data analyses,

(5) 4–6 high-quality figures,

(6) 2–4 summary tables,

(7) one full draft ready for feedback.

7 Priority Order if You Fall Behind

If you fall behind schedule, prioritize in this order:

(1) finish the simulation study,

(2) complete the methods and simulation results writing,

(3) include only two real datasets,

(4) reduce the number of figures,

(5) postpone optional extensions.

8 Final Advice

The paper will be strongest if it remains narrow, empirical, and clear. The simulation study should carry the paper, while the real-data examples should support and illustrate the main message. The goal is not to solve all of subgroup heterogeneity or all of transportability. The goal is to produce a clean, credible empirical paper with a precise question and strong execution.